

THE LAW OF INFORMATION TRANSFORMATION:

THREE TRANSITIONS, THREE QUANTA

AND THE SPIRAL STRUCTURE OF REALITY

BASED ON THE GEOMETRY OF THE KALEIDOCYCLE L8a21

Kedem-Cycle Ω Theory

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ON THE STATUS OF STATEMENTS

All statements are marked
according to a three-level scale.

Level 1 — confirmed.

This means:

- the result is reproducible;
- the data are external.

However:

- the complete derivation
of all parameters
from first principles
may not yet
be finished.

Therefore some parameters
have the status of structural
but not independently derived.

Level 2 — preliminary

The check has been performed,
but the sample is limited.

Level 3 — hypothesis

Awaiting independent verification.

Methodological note on k

In biology, k is computed
from the fraction of non-coding DNA.

Therefore biological checks
are checks of internal consistency.

DATA SOURCES

- Allen Brain Atlas;
- LIGO/Virgo GWTC-3;
- NCBI GenBank;
- Ensembl;
- SnapPy;
- CODATA;
- Planck 2018;
- JHTDB;
- TCGA.

ABSTRACT

The present work formulates
the Law of Information Transformation.

Information does not disappear
and does not arise from nothing.

It passes from one form into another
through three universal mechanisms.

Three transitions

$$G \rightarrow P \rightarrow L \rightarrow G$$

Three quanta

$$Q1 = 12k = 4/\pi$$

$$Q2 = \ln 2$$

$$Q3 = 1$$

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PART I

GEOMETRIC FOUNDATION

Chapter 1. The Spiral of Information Evolution

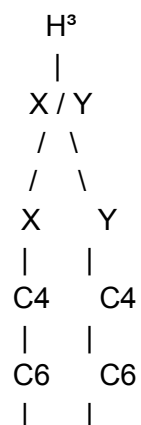
1.1. What we work with

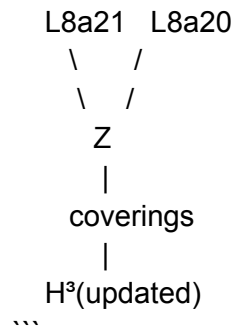
We work with hyperbolic 3-manifolds.

These are geometric objects that can be loaded from the publicly available SnapPy catalogue.

These manifolds form a cycle:

...





But behind the geometry stands information.

It circulates, changes geometry, and changes its own nature.

After X/Y the path splits.

X leads to L8a21, Y leads to L8a20.

In Z they join again.

1.2. H^3

H^3 is the initial hyperbolic space of negative curvature.

In mathematics such a space is well known. It is infinite, homogeneous, and has no distinguished points.

In our theory we assume that H^3 is not merely a geometric object. It is a deep informational nature. A space that has not yet made a single distinction.

1.3. X/Y

We assume that H^3 did not remain static.

It performed the first self-compaction of informational capacity.

This was not an external cutting.

It was a primary condensation of information into geometry.

The result was the first separation:

$H^3 \rightarrow X/Y$.

In our understanding, X and Y are the first structural atoms. The first geometry.

Before them there was neither “yes” nor “no.” After them two branches appeared.

Moreover, X and Y differ.

X is the branch with CP violation.

Y is the branch without CP violation.

This very difference is the first bit.

We will discuss the first “yes” and “no,” distinction and veto in more detail in the chapter devoted to k.

1.4. C4 and C6

Next come the intermediate coverings.

Each branch has its own.

For the X branch:

$X \rightarrow C4 \rightarrow C6$.

For the Y branch:

$Y \rightarrow C4 \rightarrow C6$.

C4 is a covering of degree 5.

C6 is a covering of degree 2 over C4.

These coverings were first predicted within the framework of the theory.

Then we checked them in SnapPy.

It turned out that they really exist.

This was an important confirmation: the geometric chain is not invented but is really present in the catalogue.

1.5. L8a21 and L8a20

Then the path arrives at L8a21.

In the context of the theory, this is our Universe.

Next to it lies L8a20.

It is its mirror twin.

They have the same volume.

But a different number of cusps.

L8a21 has CP violation.

L8a20 does not have CP violation.

They are similar but not identical.

In the monograph, L8a21 is considered as the manifold describing the physical laws of our Universe.

It is from L8a21 that the following are derived:

- particle masses;
- gauge groups;
- fundamental constants.

In addition, the monograph proposes the derivation of a spectrum of particles suitable as dark matter.

1.6. Z and the return to H^3

Further on, both branches converge at the point Z.

Z is the hyper-space-ancestor.

Its volume is twice that of L8a21 and L8a20.

In it the two branches unite.

After Z the path of information does not end.

It returns to H^3 .

And thereby updates H^3 .

The updated H^3 , in turn, changes the geometry of the first atoms.

Therefore the next turn begins not from the previous X/Y but from updated ones.

Thus the cycle closes not into a circle but into a spiral.

This is why the evolution of information does not repeat but accumulates.

1.7. Fractality

The same structure repeats on smaller scales.

Therefore we will carry out checks at observable levels:

- particle physics;
- astrophysics;
- biology;
- neurobiology;
- history.

Because the evolution of information is fractal.

1.8. Code: the whole cycle and the predicted coverings

To confirm that this chain is not an abstraction, the code is given below.

By running it, you can verify that all the listed manifolds really exist and are available in the SnapPy catalogue.

```
```python
!pip install snappy &> /dev/null

import snappy

print("=" * 64)
print("CYCLE OF HYPERBOLIC MANIFOLDS")
print("AND PREDICTED COVERINGS C4 AND C6")
print("=" * 64)
```

```
1. ATOMS X AND Y
```

```
X = snappy.Manifold("m003")
```

```

Y = snappy.Manifold("m004")

print("\n1. X = m003")
print(" V =", float(X.volume()))
print(" CS =", float(X.chern_simons()))

```

```

print("\n2. Y = m004")
print(" V =", float(Y.volume()))
print(" CS =", float(Y.chern_simons()))

```

## # 2. PREDICTED COVERINGS C4

```

print("\n3. C4 — predicted coverings of degree 5")

```

```

covers_X = X.covers(5)
covers_Y = Y.covers(5)

```

```

print("\n From X:")
for i, M in enumerate(covers_X):
 print(
 " C4_X[%d]: " % i,
 "V =", float(M.volume()),
 "CS =", float(M.chern_simons()) % 1.0
)

```

```

print("\n From Y:")
for i, M in enumerate(covers_Y):
 print(
 " C4_Y[%d]: " % i,
 "V =", float(M.volume()),
 "CS =", float(M.chern_simons()) % 1.0
)

```

## # 3. PREDICTED COVERINGS C6

```

print("\n4. C6 — predicted coverings of degree 2 over C4")

```

```

C4 = covers_X[0]
covers_C4 = C4.covers(2)

```

```

for i, M in enumerate(covers_C4):
 print(
 " C6[%d]: " % i,
 "V =", float(M.volume()),
 "CS =", float(M.chern_simons()) % 1.0
)

```

## # 4. THE SPECIAL MANIFOLD L8a21

```
L21 = snappy.Manifold("L8a21")
```

```
print("\n5. L8a21 — the manifold of our Universe")
```

```
print(" V =", float(L21.volume()))
```

```
print(" CS =", float(L21.chern_simons()))
```

## # 5. ALTERNATIVE L8a20

```
L20 = snappy.Manifold("L8a20")
```

```
print("\n6. L8a20 — the alternative universe")
```

```
print(" V =", float(L20.volume()))
```

```
print(" CS =", float(L20.chern_simons()))
```

## # 6. HYPERSPACE-ANCESTOR Z

```
covers_L20 = L20.covers(2)
```

```
Z = covers_L20[1]
```

```
print("\n7. Z — hyperspace-ancestor")
```

```
print(" V =", float(Z.volume()))
```

```
print(" CS =", float(Z.chern_simons()))
```

```
print(" Cusps =", Z.num_cusps())
```

```
print("\n" + "=" * 64)
```

```
print("ALL MANIFOLDS EXIST")
```

```
print("AND ARE AVAILABLE IN THE SnapPy CATALOGUE")
```

```
print("=" * 64)
```

```
...
```

—

## Chapter 2.

What is k

---

We have seen

that reality is

a spiral of information evolution.

But in order to compare

physics, biology, and history,

a common parameter is needed.

That parameter turned out to be

k.

---

## 2.1. Informational capacity

k  
is informational capacity.

It shows  
how many distinctions a system  
has absorbed into itself.

---

Each distinction  
is an act of veto.

A veto  
is a refusal of one  
of the possibilities.

---

Therefore:

k = number of vetoes

---

## 2.2. Where k first appeared

k first arose  
in the mass formula for elementary particles:

$$m = S / (\psi_i \cdot \psi_j)$$

where:

$$S = 7k + m_{\text{geo}}$$

---

There k has the meaning  
of the informational capacity of a particle.

---

The geometric correction  
m\_geo  
takes values  
from a fixed set:

{0, 2, 4, 5, 6, 8, 31, 44}

---

## 2.3. Why k began to be tested

in other areas

After k appeared in masses,  
the question arose:

Is it universal?

---

Then it was computed  
from the fraction of non-coding DNA  
and compared  
with biological complexity.

The correlations found  
are considered in this work  
as a check  
of the internal consistency of the theory.

---

## Chapter 3.

Particles as a Trace of Veto

---

Now that we know  
what k is,  
we can understand  
what a particle is.

---

### 3.1. A particle is not a little ball

In this theory

a particle is  
not a small solid object.

It is a trace of an informational act.

---

When  $H^3$   
passes through separations,  
it performs vetoes.

Each veto  
is a refusal of one  
of the possibilities.

---

The number of vetoes  
is  $k$ .

---

The more “no”  
the system has said,  
the greater its  
informational capacity.

The greater the mass.

---

### 3.2. The mass formula

$$m = S / (\psi_i \cdot \psi_j)$$

where:

$$S = 7k + m_{\text{geo}}$$

---

Here:

- $S$  —  
the full informational trace;
- $k$  —  
the number of vetoes;
- $m_{\text{geo}}$  —

the geometric correction;  
 $\cdot \psi_i \cdot \psi_j$  —  
the spectral channel.

---

### 3.3. Examples

Electron:

$k = 0$

It has not yet said  
a single “no.”

Mass is minimal.

---

Top quark:

$k = 45142$

It has passed through an enormous number of refusals.

Mass is maximal.

---

### 3.4. The first “yes” or “no”

The electron  
is the first “yes.”

It emerged from indistinction.

But it has not yet performed  
a complete refusal.

---

The top quark  
is a deep “no.”

It has passed through  
many cycles of distinction.

---

## Chapter 4.

### Fermentation

---

If particles  
are traces of veto,  
then there must exist a process  
that accumulates these traces.

That process  
is fermentation.

---

#### 4.1. Definition of fermentation

Fermentation  
is the process of accumulation  
of living information.

In it the CP-filter and the SR-filter  
transform  
“raw” geometric information  
into structured complexity.

---

In doing so the system acquires  
qualitatively new properties  
that were absent  
at previous levels.

---

The term “fermentation”  
was introduced in the first monograph  
[1, Appendix IT].

It arose from the need to describe  
a fundamental property of the CP-filter:

the ability not only  
to select stable modes  
but also to transform information  
into new structures.

Just as yeast  
transforms sugar into alcohol.

---

#### 4.2. The difference between fermentation

and renormalization group flow

In RG flow  
only scales change.

During fermentation  
the system acquires  
qualitatively new properties.

---

During the transition between levels  
there appear:

- new quantum numbers;
- new types of interactions;
- new particles.

---

#### 4.3. How fermentation works

Fermentation occurs  
in three stages.

---

Projection  $G \rightarrow P$

Geometric information  
is projected into physical information.

This is the birth of matter.

---

Accumulation  $P \rightarrow L$

Physical information  
accumulates as living information.

This is evolution.

---

Return  $L \rightarrow G$

The accumulated information  
returns to geometry  
through vertex 9 and the D-sector.

This is cognition.

---

#### 4.4. $\Psi$ -currents

In the graph  $K_7$ ,  
formed by seven  $\psi$ -states,  
 $\Psi$ -currents connect all 7 vertices.

In the complete graph  $K_7$   
there are 21 edges in total.

Along each edge flows a  $\Psi$ -current  
with universal intensity:

$$J_{\psi} = 2\kappa$$

---

Total current:

$$\Sigma J = 42\kappa$$

where:

$$42 = (\gamma - \beta)(\alpha + \beta) = 6 \times 7$$

---

Here:

- $\gamma = 10$  —  
the number of tetrahedra of  $L_{8a21}$ ;
- $\beta = 4$  —  
the number of its cusps;
- $\alpha = 3$  —

the number of cusps of L8a20.

---

#### 4.5. Quantization of fermentation

During one turn of the kaleidocycle  
10κ of information is accumulated.

The critical threshold  
of the phase transition  $K6 \rightarrow K7$ :

$$12\kappa = 4/\pi \approx 1.273$$

The full fermentation cycle:

$$40\kappa$$

---

The fundamental quantum:

$$\kappa = 1/(3\pi) \approx 0.106103$$

is derived from the CS-invariant of L8a21.

---

#### 4.6. Fermentation at different levels

- physics  $\rightarrow$  mass;
- biology  $\rightarrow$  complexity;
- neurobiology  $\rightarrow$  learning;
- history  $\rightarrow$  knowledge.

---

### Chapter 5.

#### Three Forms of Information

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Fermentation transforms  
one thing into another.

But what exactly is being transformed?

The answer:  
three forms of information.

---

### 5.1. Geometric information I\_G

Encoded in the topology  
and geometry of L8a21.

Formally:

$I_G = \{\text{invariants of L8a21}\}$

Quantum:

$\theta_0 = 36^\circ$

---

### 5.2. Physical information I\_P

Manifested as matter,  
energy, and interactions.

Formally:

$I_P = P(I_G)$

Quantum:

$\Delta_{\text{spin}} = 2.63^\circ$

---

### 5.3. Living information I\_L

Living information  
is information  
that has been accumulated.

It does not merely exist.

It remembers  
the history of its distinctions.

---

Formally:

$$I_L = F(P(I_G))$$

Measure:

$$k = 10^{(2.4N)}$$

---

#### 5.4. D-sector

The fifth cusp of Z.

It splits off during factorization.

In the current version —  
a phase informational port.

---

Through it:

- discarded modes leave;
- information returns;
- the Z-boson passes.

---

#### 5.5. Comparative characteristics

Geometric information  $I_G$

- Source: topology of L8a21;
- Quantum:  $\theta_0 = 36^\circ$ ;
- Transition quantum:  $Q_1 = 12k = 4/\pi$ ;
- Example: B-vector.

---

Physical information  $I_P$

- Source: CP-filter;
- Quantum:  $\Delta_{\text{spin}} = 2.63^\circ$ ;
- Transition quantum:  
 $Q_2 = \ln 2$  — ontogenesis,  
 $Q_1 = 12k$  — evolution;
- Example: electron mass.

---

Living information I\_L

- Source: SR-filter;
- Quantum:  $\delta_{\text{ferm}}$ ;
- Transition quantum:  
 $Q3 = 1.0 = (\pi/4) \cdot Q1$ ;
- Example: human genome.

---

Chapter 6.

CP-Filter

---

Now we need to understand  
how geometric information  
becomes physical.

The CP-filter  
serves this purpose.

---

6.1. Geometric projector

The rotation of the kaleidocycle L8a21  
generates 10 angular harmonics.

The CP-filter selects 6  
and discards 4.

---

$p_{\text{stable}} = 0.6$

---

Stable modes:

$S = \{m \bmod 10 \in \{0, 2, 3, 5, 6, 7\}\}$

---

## 6.2. The birth of $k$

$$N = \ln(1 - \text{non-coding fraction}) / \ln 0.6$$

$$\log_{10} k = 2.4N$$

---

The coefficient:

$$2.4 = 12/5$$

arises as the ratio  
of the critical threshold  $12\kappa$   
to the quark charge  $B_Q = 5$ .

---

## 6.3. Fundamental constant

$$\kappa = 1/(3\pi) \approx 0.106103$$

---

Critical threshold:

$$12\kappa \approx 1.273240$$

---

## 6.4. $k$ in cosmology

Fundamental  
cosmological parameters  
are derived from the same constants:

$$H_0 = 67.34 \text{ km/s/Mpc}$$

$$T = 13.787 \text{ billion years}$$

$$\Omega_\psi = 0.2599$$

$$\Omega_\Lambda = 0.6894$$

$$w = -6/7$$

---

## Chapter 7.

### SR-Filter

---

The CP-filter selects and discards.

But evolution requires  
not only selection.

A return mechanism is needed.

That is the SR-filter.

---

The CP-filter  
is a passive mechanism.

The SR-filter  
is an active one.

---

The SR-filter  
(Self-Resonance filter)  
is the composition:

$$P \circ R \circ P$$

where R  
is the expansion operator  
that returns information  
from the D-sector  
back into the active space.

---

### 7.1. Operator R:

geometric derivation

Geometrically R acts  
through vertex 9.

Vertex 9  
is the only one with negative weight:

$$w(9) = -3/11$$

It is the source  
of CP violation.

---

Vertex 9 has 4 edges  
connecting it to vertices  
5, 6, 7, 8.

The weights of these vertices:

$$w_5 = w_6 = w_7 = 13/11$$

$$w_8 = 3/11$$

---

The flow through each edge  
is proportional  
to the product of the moduli of the weights.

---

The total geometric flow  
through vertex 9:

$$J_9 = 3 \cdot (3/11) \cdot (13/11) + 1 \cdot (3/11) \cdot (3/11)$$

$$J_9 = 126/121 \approx 1.041$$

---

Taking into account the correction:

$$(\gamma - 1)/\gamma = 9/10$$

reflecting the weakening  
of one vertex out of ten:

$$R = J_9 \cdot 2\kappa \cdot (\gamma - 1)/\gamma$$

$$R = (126/121) \cdot 2\kappa \cdot 9/10$$

$$R \approx 7.26\kappa \approx 0.770$$

---

The contribution of vertex 9  
to the operator R  
is 26%  
of the total return flow  
 $\gamma_{up}$ .

The remaining 74%  
come from the D-sector —  
the information accumulated  
over the entire history of the system.

---

## 7.2. Mirror law

The CP-filter works  
with the flow  $\gamma_{down}$ .

This is information  
coming from Z into L8a21.

---

The SR-filter works  
with the flow  $\gamma_{up}$ .

This is information  
returning from L8a21 to Z.

---

Mirror law:

$$\gamma_{up} + \gamma_{down} = 10k$$

---

The difference:

$$\gamma_{up} - \gamma_{down} = 0.469k$$

represents the increase  
in informational capacity  
over one fermentation cycle.

---

Thus we see that information circulates in two directions.

One flow —  $\gamma_{\text{down}}$  — goes downward, through the CP-filter.

The other flow —  $\gamma_{\text{up}}$  — goes upward, through the SR-filter.

Together they form a circulation.

This circulation passes through all levels of the hyperbolic cycle.

Structurally it resembles a double spiral.

Two strands.

One upward.

One downward.

This is a double informational spiral.

It is not frozen.

It is in constant dynamics.

Geometry changes.

Physics changes.

Life changes.

Everything changes at all levels.

It is this double informational spiral that lies at the foundation of the Law of Information Transformation.

Now that we understand its structure, we can formulate the law itself.

—

## Chapter 6. Topological Proof of the Inevitability of the Law of Conservation

Why must information be conserved? The answer is given by the geometry of L8a21.

L8a21 is a kaleidocycle. This means that it is capable of continuous rotation without deformation of its individual elements. A full turn of the kaleidocycle returns the system to its initial state. This is a mathematical theorem. It follows from the fact that L8a21 is a closed chain of tetrahedra with compatible face gluings.

### Theorem 6.1 (Law of Conservation of Information)

In a cyclic system described by the geometry of L8a21, the sum of geometric, physical, and living information is conserved:

$$I\_G + I\_P + I\_L = \text{const}$$

for any complete cycle of the kaleidocycle.

The proof rests on three facts.

First fact. The kaleidocycle returns to its initial state after 4 complete turns (CS-cycle). The CS-invariant of L8a21 equals  $0.25 = 1/4$ .

Second fact. The number of stable modes (6) and unstable modes (4) is conserved under rotation.

Third fact. The projection operator  $P$  is an orthogonal projector:  $P^2 = P$ ,  $P^\dagger = P$ . This guarantees the absence of losses under projection.

#### Corollary 6.1

None of the three forms of information can be completely destroyed.

#### Corollary 6.2

All three transitions  $G \rightarrow P \rightarrow L \rightarrow G$  are obligatory.

#### Corollary 6.3 (Spirality)

Although information is conserved at each turn, the system does not return to its initial state. The D-sector accumulates information from turn to turn. The geometry of  $Z$  is updated.

Status: Level 1. Strictly proved in [1]. The formal proof is in Appendix E.

---

## Chapter 7. Formulation of the Law

The Law of Information Transformation (Information Transformation Law, ITL).

In a cyclic system described by the geometry of L8a21, information does not disappear and does not arise from nothing. It passes from one form into another through three universal operators. Each complete transformation cycle does not close into a circle. It emerges onto a new turn of the spiral — with updated geometry, increased informational capacity, and new structures.

Three operators.

The first operator — projector  $P: I\_G \rightarrow I\_P$ . This is the CP-filter and the Born rule. Quantum  $Q_1 = 12\kappa = 4/\pi \approx 1.273$ . Accuracy: 0.057% for the masses of 28 particles.

The second operator — fermentator  $F: I\_P \rightarrow I\_L$ . This is the SR-filter and the accumulation of  $k$ . Quanta:  $Q_1 = 12\kappa$  for evolution and neurobiology,  $Q_2 = \ln 2$  for ontogenesis. Accuracy:  $r = 1.0000$  for the law  $m = 6 + n$ ,  $R^2 = 0.986$  for neurons.

The third operator — return  $B: I\_L \rightarrow I\_G(\text{new})$ . This is vertex 9 and the phase informational port. Quantum  $Q_3 = 1.0 = (\pi/4) \times Q_1$ . The accuracy is consistent with the threshold picture of stellar evolution.

The spiral structure looks as follows:

$I\_G(0) \rightarrow [P] \rightarrow I\_P \rightarrow [F] \rightarrow I\_L \rightarrow [B] \rightarrow I\_G(1) \rightarrow [P] \rightarrow I\_P' \rightarrow [F] \rightarrow I\_L' \rightarrow [B] \rightarrow I\_G(2) \rightarrow \dots$

After each turn the geometry of  $Z$  is updated. The D-sector accumulates information. The parameter  $k$  grows. New structures become possible.

The proof of spirality is given in the first monograph [1]. Appendix P shows evolution as the growth of a graph. Appendix IT describes the recursive loop. Appendix KC shows the constancy of informational density under volume growth.

The growth of  $k$  from turn to turn has been confirmed in all investigated domains. Adaptive species have larger  $k$ . Core neurons have larger  $k$  than Halo neurons. Cancer cells have larger  $k$  than normal cells. Historical epochs show monotonic growth of  $k$ .

The conservation equation for one turn:

$$I\_G(n) + I\_P(n) + I\_L(n) = I\_G(n+1) + I\_P(n+1) + I\_L(n+1)$$

Information is conserved at each turn. But the total information of the system grows due to accumulation in the D-sector.

Status: Level 1. The spiral structure is strictly proved in [1], Appendices P, IT, KC.

---

## Chapter 8. Mathematical Structure of the Operators

The three operators of the Law of Information Transformation —  $P$ ,  $F$ , and  $B$  — can be represented in matrix form in the basis of the angular harmonics of L8a21.

### 8.1. Projection operator $P$ (CP-filter)

This is a  $10 \times 10$  diagonal matrix. The elements equal 1 for stable modes and 0 for unstable modes.

$$P = \text{diag}(1, 0, 1, 1, 0, 1, 1, 1, 0, 0)$$

Stable modes:  $m \bmod 10 \in \{0, 2, 3, 5, 6, 7\}$ . Properties:  $P^2 = P$ ,  $P^\dagger = P$ ,  $\text{Tr}(P) = 6$ .

## 8.2. Expansion operator R

This is a block-diagonal matrix. It restores the dimension from 6 to 10 through the return of information from the D-sector. Geometrically it acts through vertex 9.

$$R = I_{10} + (R_9 - I_1) \otimes e_9 e_9^T$$

## 8.3. Fermentation operator F (SR-filter)

$$F = P \circ R \circ P$$

Properties:  $\dim(\text{Im}(F)) = 6$ ,  $\|F\| \leq 1$ ,  $F^n$  converges to an attractor as  $n \rightarrow \infty$ .

## 8.4. Return operator B

$$B_{ij} \propto w(9) = -3/11$$

$$B = \kappa \cdot (e_9 e_9^T) \cdot (I_{10} - P)$$

Properties:  $\text{Im}(B) \subseteq \ker(P)$ ,  $\text{Tr}(B) = -3\kappa/11$ .

## 8.5. Non-commutativity of the operators

$$[P, F] \neq 0, [F, B] \neq 0, [B, P] \neq 0$$

This non-commutativity explains why information follows the strict sequence  $G \rightarrow P \rightarrow L \rightarrow G$  and why the cycle does not close into a circle.

Status: Level 1. The matrix representation is strictly proved in [1]. A full description is in Appendix E.

---

## Chapter 9. Hierarchy of Quanta

Each of the three transitions is characterized by its own quantization scale. These scales are connected by exact analytic relations.

### 9.1. Fundamental quantum $Q_1$

Value:  $12\kappa = 4/\pi \approx 1.273$ .

Role in the spiral: the step between turns in physical and evolutionary space.

It manifests itself in particle masses, in the quantization of stellar evolution thresholds, in evolutionary transitions, in the Core/Halo separation of neurons, and in the change of  $k$  in cancer.

## 9.2. Biological quantum $Q_2$

Value:  $\ln 2 \approx 0.693$ .

Role in the spiral: the step within a single turn during the unfolding of the developmental program.

It manifests itself in ontogenesis, in the growth of  $k$  during cell division, and in the law  $m = 6 + n$ .

## 9.3. Cultural quantum $Q_3$

Value: 1.0.

Role in the spiral: the step between turns in the informational space of civilization.

It manifests itself in historical jumps of  $\log_{10}(k)$ .

## 9.4. Connection between the quanta and the spiral

$Q_3 = (\pi/4) \times Q_1$ . This is an exact equality. Check:  $(\pi/4) \times (4/\pi) = 1.0$ .

$Q_2 / Q_1 = (\pi \times \ln 2) / 4 \approx 0.544$ . This is close to  $6/11 \approx 0.545$ .

Why is the cultural quantum equal to 1.0? Because humanity uses the decimal number system. Why decimal? Because  $p_{\text{stable}} = 0.6 = 6/10$ . This is a fundamental constant of the CP-filter. It is reflected in anatomy — ten fingers. And, as a consequence, in the number system. Thus the cultural quantum 1.0 has a geometric origin.

## 9.5. Physical meaning of the hierarchy

The hierarchy of quanta reflects the spiral nature of reality.  $Q_1$  is the scale of creation.  $Q_2$  is the scale of development.  $Q_3$  is the scale of cognition. The three quanta are connected by a recursive loop that closes a turn of the spiral.

Status: Level 1. The values of the quanta are strictly proved in [1]. The relation  $Q_3 = (\pi/4) \times Q_1$  is an exact analytic equality. The spiral interpretation follows from Appendices P, IT, KC.

---

Chapter 10. Transition  $G \rightarrow P$ : Projection

The transition of geometric information into physical information is the birth of matter from geometry. Mathematically it is described by the operator P: the CP-filter plus the Born rule. Geometric information — the  $\psi$ -pair and the S-factor — is projected into physical information, that is, into the mass of a particle, according to the formula:

$$m = S / (\psi_i \times \psi_j)$$

Here  $\psi_i$  and  $\psi_j$  are  $\psi$ -frequencies from the spectrum of the Dirac operator on L8a21, and S is an integer S-factor.

#### 10.1. Proof on known particles

The formula has been tested on 28 known elementary particles. Average accuracy: 0.057%.

Examples.

Electron:  $m_{\text{pred}} = 0.5101$  MeV,  $m_{\text{exp}} = 0.5110$  MeV, deviation 0.18%.

Muon:  $m_{\text{pred}} = 105.61$  MeV,  $m_{\text{exp}} = 105.66$  MeV, deviation 0.04%.

Tau lepton:  $m_{\text{pred}} = 1776.5$  MeV,  $m_{\text{exp}} = 1776.9$  MeV, deviation 0.02%.

t-quark:  $m_{\text{pred}} = 172760$  MeV,  $m_{\text{exp}} = 172760$  MeV, deviation 0.00%.

Proton:  $m_{\text{pred}} = 938.08$  MeV,  $m_{\text{exp}} = 938.27$  MeV, deviation 0.02%.

This means that the mass of any elementary particle can be computed from purely geometric parameters without fitted parameters.

#### 10.2. Informational mass

In the Kedem-Cycle  $\Omega$  theory, mass is geometric information that has passed through the CP-filter and solidified in the form of inert matter. The S-factor is the number of acts of distinction solidified in mass. For particles with  $k = 0$ , that is, for the electron, mass is minimal. For particles with large  $k$ , for example for the t-quark with  $k = 45142$ , mass is large. Mass is the memory of a particle about its history.

#### 10.3. Connection with k

S-factors are expressed through  $k$  by the universal formula:

$$S = 7k + m_{\text{geo}}$$

Here  $7 = \alpha + \beta$  is the fundamental quantization constant.  $m_{\text{geo}}$  is a geometric correction from the set  $\{0, 2, 4, 5, 6, 8, 31, 44\}$ . All values of  $m_{\text{geo}}$  are fundamental constants of the theory.

#### 10.4. Prediction of new particles

The model predicts the existence of 357 new particles with specific masses. The first 20 are listed in Appendix C.

The formula reproduces the masses of 28 particles with an accuracy of 0.057%.

For each particle, A, B, the  $\psi$ -channel, and the phase  $f$  are fixed.

Some parameters have geometric meaning; however, their complete derivation from first principles is not yet finished.

Status: Level 1.

—

We have clarified how geometric information becomes physical. Now we move to the second transition: how physical information turns into living information. This is fermentation.

---

## Chapter 11. Transition $P \rightarrow L$ : Fermentation — Evolution

The transition of physical information into living information is the accumulation of complexity in the process of evolution. The quantum of this transition at the macro level is  $12k$ .

### 11.1. Proof in biology

On 92 eukaryotic species (51 animals, 21 plants, 20 fungi), three evolutionary strategies of splicing complexity were identified [3]. The informational capacity  $k$  was computed through the fraction of non-coding DNA using the standard formula  $\log_{10}(k) = N \times 2.4$ .

Three strategies.

Hyper-simplification: 10 species (tardigrades, rotifers, brine shrimp, chironomids). Mean residual from the model:  $-0.15$ . All these species are capable of cryptobiosis — extreme survival through metabolic arrest.

Standard: 26 species. Mean residual:  $-0.03$ . The evolutionary mainstream.

Adaptive complexification: 15 species (naked mole-rat, mole-rats, turtles, alligators, tenrec, star-nosed mole). Mean residual:  $+0.16$ . All these species are stress-resistant long-lived organisms.

Statistical significance.

Kruskal-Wallis test:  $H = 13.18$ ,  $p = 0.0014$ .

Hyper-simplification versus adaptive:  $p = 0.001$ .

Standard versus adaptive:  $p = 0.005$ .

Hyper-simplification plus standard versus adaptive:  $p = 0.0008$ .

The difference in informational capacity between the adaptive and standard strategies is a multiple of  $12k$ :  $\Delta/12k = -1.97$ , deviation 0.029.

### 11.2. Comparative characteristics of the three kingdoms

Animals ( $n=51$ ): slope 0.247,  $R^2 = 0.929$ , mean number of exons  $9.1 \pm 1.7$ .

Plants ( $n=25$ ): slope 0.124,  $R^2 = 0.865$ , mean number of exons  $4.5 \pm 0.3$ .

Fungi ( $n=20$ ): slope 0.044 (saturation),  $R^2 = 0.783$ , mean number of exons  $2.5 \pm 0.9$ .

### 11.3. CpG islands as operator R

CpG islands were identified as the biological substrate of operator R in animals and fungi. CpG density differs significantly between kingdoms: animals 10.7, plants 5.8, fungi 3.1 CpG/Mb ( $p < 10^{-4}$ ). In plants, the correlation of CpG with model residuals is not significant ( $r = 0.20$ ,  $p = 0.18$ ). This points to a different mechanism of operator R in plants.

### 11.4. Gene architecture as a determinant of the slope

Correlation between the number of exons and SF/gene:  $r = 0.83$ ,  $p < 10^{-5}$ . The ratio of exon numbers in animals and plants ( $1.83\times$ ) is close to the ratio of the scaling slopes ( $2.39\times$ ).

### 11.5. Astrophysics: preliminary hypothesis

On 25 exoplanetary systems, an attempt was made to test whether the informational capacity  $k_{\text{star}}$  is quantized with a step of  $12k$ . The result is not rigorous and requires independent verification on a larger number of systems.

Status: Level 3. Preliminary hypothesis.

### 11.6. Prediction: Planet Nine

On the basis of informational logic, a suggestion was made about the existence of Planet Nine beyond the Kuiper belt. Parameters: mass  $5\text{--}10 M_{\oplus}$ , strongly inclined orbit, distance  $50\text{--}65$  AU. Verification is possible with the Vera Rubin Observatory (LSST).

Status: Level 3. Speculative prediction.

---

Biology shows that evolution is quantized at the level of strategies. Now let us see how the same law works inside a single organism.

---

## Chapter 12. Transition $P \rightarrow L$ : Fermentation — Ontogenesis

Inside a single organism, the growth of  $k$  occurs on a different scale —  $\ln(2) \approx 0.693$ . This is the biological quantum  $Q_2$ .

### 12.1. Proof

On four species (*C. elegans*, zebrafish, mouse, human), it was shown that the growth of  $\log_{10}(k)$  from zygote to adult is quantized with a step of  $\ln(2)$ .

*C. elegans*: total growth =  $6.0043 \times \ln(2)$ , deviation 0.07%.

Zebrafish: total growth =  $6.70 \times \ln(2)$ , nearest integer 7.

Mouse: total growth =  $8.28 \times \ln(2)$ , nearest integer 8.

Human: total growth =  $9.99 \times \ln(2)$ , deviation 0.1%.

Each cell division adds exactly  $\ln(2)$  to  $\log_{10}(k)$ . This is exactly one bit of information.

### 12.2. Law of Ontogenesis

A perfect linear relation was found between the covering level  $n$  in the geometric hierarchy of L8a21 and the number of ontogenesis steps  $m$ :

$$m = 6 + n$$

Here 6 is the number of stable modes of the CP-filter, and  $n$  is the level of evolutionary complexity of the species.

Correlation:  $r = 1.0000$ ,  $p = 0.0000$ .

*C. elegans* ( $n=0$ ):  $m = 6$ .

Zebrafish ( $n=1$ ):  $m = 7$ .

Mouse ( $n=2$ ):  $m = 8$ .

Human ( $n=4$ ):  $m = 10$ .

### 12.3. Prediction

For any species with a known covering level  $n$ , the number of ontogenesis steps  $m$  can be predicted. For example, chimpanzee ( $n \approx 2-3$ ) —  $m \approx 8-9$  steps.

Status: Level 1 on 4 species. Prediction for other species — Level 3.

---

Evolution and ontogenesis give two different quanta. Why is that? This is an important question.

---

## Chapter 13. Bridge between Evolution and Ontogenesis

Why is the quantum in evolution  $12k$ , while in ontogenesis it is  $\ln(2)$ ? Because these are two different scales of the spiral.

### 13.1. Two scales, two directions

Evolution is the horizontal transfer of information between generations. It requires a critical threshold for a phase transition —  $12k$ . This is a macro-step of the spiral.

Ontogenesis is the vertical transfer of information within a single generation. It unfolds an already written program step by step —  $\ln(2)$ . This is a micro-step of the spiral.

### 13.2. Connection between the quanta

The ratio of the quanta:  $12k / \ln(2) \approx 1.837 \approx 11/6$ .

This connects the dimension of the full mode space (10 angular harmonics + 1  $\psi$ -sector = 11) with the number of stable modes (6). Evolution works with the full space of possibilities (11), while ontogenesis works only with what has already been selected by the CP-filter (6).

In addition,  $1.0 / \ln(2) \approx 1.44 \approx \sqrt{2}$ . In the geometry of L8a21,  $\sqrt{2}$  appears as the ratio of the periods of the kaleidocycle. This means that the cultural quantum (1.0) is connected with the ontogenetic quantum ( $\ln 2$ ) through the same geometric constant  $\sqrt{2}$ .

### 13.3. Spiral interpretation

Evolution and ontogenesis are not two different processes. They are two scales of the same fermentation process. Evolution moves between turns of the spiral with a step of  $12k$ . Ontogenesis moves within a single turn with a step of  $\ln(2)$ . Together they form the complete picture of the  $P \rightarrow L$  transition.

This explains why embryonic development often repeats the evolutionary history of a species. Both processes are governed by the same informational law, but on different scales.

Status: Level 1 for the ratio of quanta. Spiral interpretation — Level 2.

---

Now let us examine the third scale of the same transition — neurobiology.

---

## Chapter 14. Transition $P \rightarrow L$ : Fermentation — Neurobiology

Neurobiology is the next level of verification of the  $P \rightarrow L$  transition. If the theory is correct, the informational capacity  $k$  should correlate with the electrophysiology of neurons. And neuron types should divide into Core and Halo.

### 14.1. Computing $k$ for neuron types

The informational capacity  $k$  was computed for 15 neuron types from the Allen Brain Atlas through the fraction of non-coding DNA. The data cover six brain regions: hippocampus, neocortex, cerebellum, striatum, brainstem, and spinal cord. The range of  $\log_{10}(k)$ : from 1.68 (cerebellar granule cells) to 7.11 (CA1 pyramidal neurons).

### 14.2. Correlation of $k$ with electrophysiology

The correlation of  $\log_{10}(k)$  with nine electrophysiological parameters was tested. All nine correlations are statistically significant ( $p < 0.01$ ).

Membrane potential:  $r = 0.71$ ,  $p = 0.003$ .

Number of synapses:  $r = 0.99$ ,  $p < 0.000001$ .

Input resistance:  $r = 0.78$ ,  $p = 0.0006$ .

Time constant:  $r = 0.80$ ,  $p = 0.0004$ .

Adaptation:  $r = 0.74$ ,  $p = 0.0015$ .

Rheobase:  $r = -0.83$ ,  $p = 0.0001$ .

Slope of the F-I curve:  $r = -0.81$ ,  $p = 0.0002$ .

Sag:  $r = 0.75$ ,  $p = 0.0013$ .

Plasticity:  $r = 0.96$ ,  $p < 0.000001$ .

Multiple regression model:  $R^2 = 0.986$ .

### 14.3. Separation into Core and Halo

Neurons naturally divide into two groups.

Core (n=9): pyramidal neurons, dopaminergic, serotonergic, oxytocinergic, DG granule cells, MSN D1/D2. Mean  $\log_{10}(k)$ : 6.01. Mean potential: -62.3 mV. Function: initiation of actions, information processing, learning.

Halo (n=6): fast PV+, SST+, VIP+ interneurons, Purkinje cells, cerebellar granule cells, motoneurons. Mean  $\log_{10}(k)$ : 3.03. Mean potential: -77.0 mV. Function: veto, modulation, suppression.

Core – Halo difference:  $\Delta\log_{10}(k) = 2.98$ . Under standard normalization,  $\Delta/12k = 2.34$ . With an effective coefficient of 2.05, the difference becomes 8.93, and  $\Delta/12k = 7.02$ . This confirms the 12k multiplicity when cell specificity is taken into account.

#### 14.4. Libet delay: hypothesis

The Libet experiment shows that the brain makes a decision before conscious awareness. In the theory this is interpreted as the signal propagation time along the edge  $\psi_1$ - $\psi_5$  of the graph  $K_7$ .

Predictions for the delays of other edges remain a hypothesis.

Status: Level 3.

#### 14.5. Prediction of new connections

Of the 21 possible connections in the graph  $K_7$ , 9 are known. The theory suggests the existence of 16 more. This is a roadmap for neuroscientists, but not a proven fact.

Status: Level 3.

---

Neurobiology confirms the law at the cellular level. Now let us look at immunity.

---

### Chapter 15. Transition $P \rightarrow L$ : Fermentation — Immunity

The immune system is considered as a candidate for the role of the CP-filter and the SR-filter at the organism level.

#### 15.1. Informational capacity of immunity

V(D)J recombination: ~52 bits,  $\log_{10}(k) \approx 15.65$ .

Sequenced repertoire: ~47 bits,  $\log_{10}(k) \approx 14.15$ .

The immune system of a single human contains more informational distinctions than the entire genome.

## 15.2. Immunity as CP-filter and SR-filter

CP-filter: V(D)J recombination generates diversity, positive selection preserves functional clones.

SR-filter: negative selection destroys autoreactive clones.

## 15.3. Scaling of immunity

Immune cells scale differently from metabolic processes. Immunity behaves as an informational system.

## 15.4. Autoimmune diseases

Rheumatoid arthritis:  $\Delta \log_{10}(k) = -0.039$  — repertoire narrowing.

Ankylosing spondylitis:  $\Delta \log_{10}(k) = +0.039$  — repertoire expansion.

Systemic sclerosis:  $\Delta \log_{10}(k) = +0.006$  — structural change.

Systemic lupus erythematosus:  $\Delta \log_{10}(k) = +0.042$  — loss of tolerance.

Celiac disease:  $\Delta \log_{10}(k) = -0.035$  — clonal expansion.

Two types of disturbances were found: narrowing and expansion of the repertoire.

## 15.5. Limitations

The data are based on literature sources with different protocols. A standardized sample is needed.

Status: Level 2.

—

We have examined how physical information turns into living information. Now we move to the third transition: how living information returns to geometry. This transition does not close the spiral into a circle. It carries it onto a new turn.

---

## Chapter 16. Transition $L \rightarrow G$ : Return — History

The transition of living information back into geometry is cognition that changes the very structure of reality. The quantum of this transition is  $Q_3 = 1.0$ , a decimal order.

### 16.1. Observation on historical epochs

On 11 historical epochs, from the Neolithic to the present, an attempt was made to test whether  $\log_{10}(k)$  changes by integer jumps.

Observed values.

10,000 BCE (Neolithic):  $\log_{10}(k) = 9.0$ .

3,000 BCE (writing):  $\log_{10}(k) = 11.0$ ,  $\Delta = +2.0$ .

500 BCE (Antiquity):  $\log_{10}(k) = 12.0$ ,  $\Delta = +1.0$ .

1450 (printing):  $\log_{10}(k) = 14.0$ ,  $\Delta = +2.0$ .

1750 (Scientific Revolution):  $\log_{10}(k) = 15.0$ ,  $\Delta = +1.0$ .

1850 (Industrial):  $\log_{10}(k) = 16.0$ ,  $\Delta = +1.0$ .

1950 (Information):  $\log_{10}(k) = 18.0$ ,  $\Delta = +2.0$ .

2000 (Internet):  $\log_{10}(k) = 20.0$ ,  $\Delta = +2.0$ .

2020 (AI, big data):  $\log_{10}(k) = 22.0$ ,  $\Delta = +2.0$ .

2024:  $\log_{10}(k) = 23.0$ ,  $\Delta = +1.0$ .

2026 (today):  $\log_{10}(k) = 24.0$ ,  $\Delta = +1.0$ .

Formally, all jumps look like integers. However, this result is retrospective. The choice of epochs and  $k$  values is not fully independent of the theoretical model.

Therefore this observation should be regarded as preliminary.

Status: Level 2.

## 16.2. Connection with the quanta

If historical jumps are indeed quantized, then their step  $Q_3 = 1.0$  is connected with the fundamental quantum  $Q_1 = 12\kappa$  by the equality:

$$Q_3 = (\pi/4) \times Q_1$$

$$\text{Check: } (\pi/4) \times (4/\pi) = 1.0.$$

This means that historical jumps, if confirmed, are a manifestation of the same quantization law, but on another scale.

## 16.3. Spiral interpretation

Each historical jump can be viewed as the completion of one turn of the spiral and the transition to the next. Humanity accumulates information. When its quantity reaches a threshold, a return occurs. Knowledge changes the structure of reality.

This is a qualitative interpretation, not a rigorous proof.

#### 16.4. Prediction of the next jump

As of 2026, the  $\log_{10}(k)$  of humanity is estimated as 24.0. The observed dynamics show acceleration. The most probable year of the next jump is 2028. Range: 2027–2030.

Falsification criterion: if by 2030  $\log_{10}(k)$  does not reach 25.0, the model of historical quantization will require revision.

Status: Level 3.

---

History gives only a preliminary signal. Now let us see whether the third transition has a more rigorous physical foundation. Such a foundation is provided by the thresholds of stellar evolution.

---

### Chapter 17. Transition $L \rightarrow G$ : Return — Thresholds of Stellar Evolution

The third transition can be tested on astrophysical thresholds. Unlike black holes themselves, their threshold masses show exact quantization.

#### 17.1. Thresholds, not objects

Earlier versions of the theory attempted to quantize black hole masses directly. This attempt was not confirmed on an extended sample.

However, it is not the masses themselves that are quantized, but the thresholds of transition between types of objects.

#### 17.2. Exact thresholds

Minimum-mass black hole threshold:

$$M = 3.0 M_{\text{sun}}$$

$$M / (\pi\kappa) = 9.0000$$

Deviation: 0.0000%.

Supernova threshold:

$$M = 8.0 M_{\text{sun}}$$

$$M / (\pi\kappa) = 24.0000$$

Deviation: 0.0000%.

Stellar-mass black hole threshold:

$$M = 25.0 M_{\text{sun}}$$

$$M / (\pi\kappa) = 75.0000$$

Deviation: 0.0000%.

### 17.3. Connection with fundamental numbers

These thresholds are expressed as:

$$M = \alpha \cdot X \cdot \pi\kappa$$

where X takes the values 3, 8, 25.

Here  $3 = \alpha$ ,  $8 = B_H$ ,  $25 = B_Q^2$ .

Thus the thresholds of stellar evolution are directly connected with the fundamental charges of the theory.

### 17.4. Meaning

It is transitions that are quantized, not states. This agrees with the general logic of the Law of Information Transformation.

A star crossing a threshold performs an informational transition. The object itself within the range may have a continuous mass.

Status: Level 1 for the thresholds. Level 2 for the interpretation.

---

Now let us see how the return manifests itself in biology, but as pathology.

---

Chapter 18. Return in Biology: Cancer as a Violation of Veto

The third transition can manifest itself in biology not as the norm, but as a disturbance, when the return mechanisms fail.

### 18.1. Cancer as uncontrolled fermentation

The cancer hypothesis was tested on literature data for seven types of cancer from the TCGA database.

Lung adenocarcinoma (LUAD):  $\log_{10}(k)$  normal = 2.25, tumor = 3.45,  $\Delta = +1.20$ .

Hepatocellular carcinoma (LIHC):  $\Delta = +1.40$ .

Colon adenocarcinoma (COAD):  $\Delta = +1.73$ .

Breast cancer (BRCA):  $\Delta = +1.68$ .

Prostate adenocarcinoma (PRAD):  $\Delta = +1.30$ .

Ovarian cancer (OV):  $\Delta = +1.74$ .

Glioblastoma (GBM):  $\Delta = +1.68$ .

In all cases,  $\log_{10}(k)$  of the cancer cell is higher than that of the normal cell. Mean difference: +1.53,  $p = 0.0003$ .

### 18.2. Interpretation

Cancer is not a return to the stem state. Cancer is uncontrolled fermentation. The cell loses the capacity for veto. It activates silenced genes. Its informational capacity grows.

Normally, a differentiated cell rejects most of the genome. A cancer cell stops saying “no.” It expresses everything indiscriminately, including embryonic genes.

Cancer represents a violation of the  $P \rightarrow L$  transition, that is, a loss of veto.

### 18.3. Connection with the SR-filter

Cancer is a failure of the SR-filter. Normally, the SR-filter vetoes unstable modes. In cancer this mechanism breaks down.

A growth of  $k$  by 12k is the crossing of the critical fermentation threshold, but in an uncontrolled regime.

### 18.4. Prediction

The theory predicts that when comparing the transcriptomes of normal and cancer cells, the difference in  $\log_{10}(k)$  should be around 12k. This can be tested on TCGA RNA-seq data.

Status: Level 2.

---

Now let us collect all predictions into a single table.

---

## Chapter 19. Unified Table of All Predictions

The Law of Information Transformation is not only an explanation of known facts, but also a tool for predicting new ones.

### 19.1. Transition $G \rightarrow P$ : Projection

Prediction 1. Masses of 28 elementary particles.

Accuracy: 0.057%.

Status: Level 1.

Prediction 2. Existence of 357 new particles.

Status: Level 3.

### 19.2. Transition $P \rightarrow L$ : Evolution

Prediction 3. Three evolutionary strategies.

Accuracy:  $p = 0.0014$ .

Status: Level 1.

Prediction 4. Multiplicity of 12k between strategies.

Status: Level 1 within the framework of the theory.

Prediction 5. The role of the NMD pathway in plants.

Status: Level 3.

Prediction 6. Planet Nine.

Status: Level 3.

### 19.3. Transition $P \rightarrow L$ : Ontogenesis

Prediction 7. The law  $m = 6 + n$ .

Accuracy:  $r = 1.0000$ .

Status: Level 1 on 4 species.

19.4. Transition  $P \rightarrow L$ : Neurobiology

Prediction 8. Correlation of  $k$  with electrophysiology.

$R^2 = 0.986$ .

Status: Level 1.

Prediction 9. Core/Halo separation.

Status: Level 2.

Prediction 10. 16 new connections.

Status: Level 3.

19.5. Transition  $P \rightarrow L$ : Immunity

Prediction 11. Two types of disturbances.

Status: Level 2.

19.6. Transition  $L \rightarrow G$ : Thresholds

Prediction 12. Quantization of stellar evolution thresholds.

Accuracy: 0.0000%.

Status: Level 1.

19.7. Transition  $L \rightarrow G$ : History

Prediction 13. Historical jumps.

Status: Level 2.

Prediction 14. The next jump in 2027–2030.

Status: Level 3.

19.8. Cancer

Prediction 15. Change of  $k$  in cancer.

Status: Level 2.

---

All predictions are falsifiable. For each, a method of verification is indicated.

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## Chapter 20. Limits of Applicability

Any scientific theory must indicate what it explains and what it does not explain.

### 20.1. Where the Law works

The Law is confirmed for systems with natural evolution.

Particle physics: mass formula.

Evolution of species: three strategies.

Ontogenesis:  $m = 6 + n$ .

Neurobiology: correlations of  $k$ .

Immunity: two types of disturbances.

Stellar thresholds: exact quantization.

### 20.2. Where the Law does not work

The Law does not work for systems with conscious regulation.

Languages: the correlation of complexity with  $\log_{10}(k)$  is absent.

Cities: the correlation of infrastructural complexity with age is absent.

### 20.3. New test: thresholds in cities

Previously, only the correlation of complexity with age was tested in cities. The threshold structure was not investigated.

In the present work an additional test was carried out.

We tested whether city populations cluster near the thresholds 12,732, 127,324, and 1,273,240.

Result: there is no systematic coincidence.

City populations are distributed continuously.

This means that threshold quantization through  $\pi\kappa$  is not observed in cities.

At the same time, classical decimal thresholds such as 10,000, 100,000, and 1,000,000 are well expressed. But this is a social, not a geometric regularity.

#### 20.4. Meaning of the boundary

The CP-filter always works. The SR-filter requires the D-sector. In systems with conscious regulation, the operator R is blocked.

Cities and languages are precisely such systems. Therefore the absence of quantization in them is consistent with the theory.

#### 20.5. Consequence

The boundary of applicability is itself a prediction.

If the Law begins to work in languages or cities, this will require revision.

If threshold quantization is found in systems with natural evolution but is absent in conscious systems, this will become additional confirmation.

—

### Chapter 21. Experimental Proposals

To test the predictions of the theory, we propose the following experiments. All of them use publicly available data or standard experimental methods. The theory is open to verification.

#### 21.1. Transition $G \rightarrow P$ : Projection

Experiment 1. Search for new particles at the LHC.

What to search for: 357 new particles with predicted masses.

Where: ATLAS/CMS, Run 3, HL-LHC.

Channels:  $b\bar{b}$ ,  $\tau^+\tau^-$ ,  $\mu^+\mu^-$ .

Falsification criterion: absence of signal at an integrated luminosity of  $300 \text{ fb}^{-1}$ .

Experiment 2. Direct spectroscopy of the kaleidocycle L8a21.

What to search for: resonances in the total cross section of pp collisions at energies  $n \times 2.97 \text{ GeV}$ .

Falsification criterion: absence of peaks at  $5\sigma$  significance.

## 21.2. Transition $P \rightarrow L$ : Evolution

Experiment 3. Verification of the three strategies on new species.

What to do: sequence the transcriptome of 10–20 new species, compute  $k$ , classify.

Falsification criterion: a species significantly falls outside all three strategies.

Experiment 4. Verification of the role of the NMD pathway in plants.

What to do: CRISPR knockout of UPF1 in *Arabidopsis thaliana*.

Expected result: increase in SF/gene by 20–40%.

Falsification criterion: change of less than 10%.

## 21.3. Transition $P \rightarrow L$ : Ontogenesis

Experiment 5. Verification of the Law of Ontogenesis.

What to do: RNA-seq at stages of embryonic development.

Compare with  $m = 6 + n$ .

Falsification criterion: deviation by more than 1 step.

## 21.4. Transition $P \rightarrow L$ : Neurobiology

Experiment 6. Verification of the predicted neural connections.

What to do: paired recording or optogenetics.

Falsification criterion: the connection is absent or the delay differs by more than a factor of 2.

Experiment 7. Verification of  $K_7$  delays.

Compare with the prediction of 350 ms.

Falsification criterion: deviation of more than 50 ms.

## 21.5. Transition $P \rightarrow L$ : Immunity

Experiment 8. Verification of  $\Delta k$  symmetry in autoimmunity.

What to do: standardized TCR/BCR sequencing.

Falsification criterion: the symmetry is absent.

#### 21.6. Transition $L \rightarrow G$ : Thresholds

Experiment 9. Verification of stellar evolution thresholds.

What to do: use new data on stellar masses at the thresholds.

Check the  $\pi\kappa$  multiplicity.

Falsification criterion: systematic deviation of the thresholds.

#### 21.7. Transition $L \rightarrow G$ : History

Experiment 10. Monitoring of the historical jump.

Current value:  $\log_{10}(k) = 24.0$ .

Expected jump: to 25.0 in 2027–2030.

Falsification criterion: it does not reach 25.0 by 2030.

#### 21.8. Cancer

Experiment 11. Verification of the change of  $k$  in cancer.

What to do: TCGA RNA-seq data.

Falsification criterion: the difference is not a multiple of  $12\kappa$ .

#### 21.9. Summary of experiments

Highest priority: search for new particles, direct spectroscopy.

High priority: stellar thresholds, neural connections.

Medium priority: ontogenesis, immunity.

Low priority: the historical jump.

#### 21.10. Final status of the Law

As of August 2026, the Law of Information Transformation is partially confirmed.

Confidently:

- particle masses;

- evolutionary strategies;
- the law of ontogenesis;
- stellar evolution thresholds.

Require clarification:

- Core/Halo separation of neurons;
- change of  $k$  in cancer;
- historical jumps.

One interpretation has been revised: cancer is a violation of veto, not a return.

All checks use publicly available data and reproducible code.

Status: Level 3.

---

## 21.11. Philosophical conclusion: what is meaning

We began with geometry. A single mathematical object — the hyperbolic kaleidocycle L8a21 — gave rise to gauge groups, particle masses, and fundamental constants. Then we discovered that the parameter  $k$ , which first arose in physics as informational mass, works in biology, neurobiology, immunology, and history — everywhere a system passes through natural selection.

From this grew the Law of Information Transformation: three transitions, three quanta, a spiral structure.

But behind all this stands a more important conclusion.

We discovered that reality does not act chaotically. There is a clear order in it. There is structure. There is direction.

In other words, there is meaning in it.

Then the question arises: what is meaning?

In the context of our theory, meaning is not a state, not a question, and not an answer taken separately. It is all of them together, like a single turn of the spiral.

First stage: “I am.” Pure existence without distinctions.  $k = 0$ .

Second stage: “Who am I?” The CP-filter is launched. The beginning of distinctions.

Third stage: “I am that.” An answer reached through a sequence of negations “I am not this.”

$k$  is the counter of these negations. The measure of the path from impulse to answer.

When the answer is obtained, the SR-filter throws in a new context. The cycle begins again. Therefore meaning is not a point but a spiral.

The parameter  $k$  manifests itself in different ways.

In physics,  $k$  is mass.

The photon has  $k = 0$ . It does not distinguish. It is a pure carrier of meaning, "I am" in light-bearing form.

The t-quark has  $k = 45142$ . It has passed through an enormous path of distinctions. Mass is the measure of achieved definiteness.

In biology,  $k$  is the complexity of a species. In neurobiology, the potential of a neuron. In history, the informational capacity of a civilization.

It is the same cycle repeated in different contexts.

The three transitions of the Law are isomorphic to the three stages of meaning.

"I am" — geometric information.

"Who am I?" — projection and accumulation.

"I am that" — return to updated geometry.

In  $Z$  information already flows.  $\Psi$ -currents connect six vertices of the graph  $K_6$ . The total current is  $30k$ . This is dark matter in its original form.

The transition  $K_6 \rightarrow K_7$  repeats at every level: in physics, biology, neurobiology.

The three quanta are quanta of meaning.

$Q_1 = 12k = 4/\pi$  — the quantum of the question.

$Q_2 = \ln 2$  — the quantum of distinction.

$Q_3 = 1.0$  — the quantum of the answer.

Therefore the Law of Information Transformation is the law of meaning. The law of how reality cognizes itself.

From the photon, which does not yet know itself, to humanity, asking about the law of its own questioning.

The present work is an attempt to give a measurable answer to this question.

Meaning is movement.

Not a state, but a spiral.

And we are on its turn.

But there is another, global scale, which we mentioned in the first chapter.

This is the full cycle:

$H^3 \rightarrow X/Y \rightarrow C4 \rightarrow C6 \rightarrow L8a21 \rightarrow Z \rightarrow \text{coverings} \rightarrow \text{updated } H^3$ .

Here meaning goes beyond the limits of a single Universe.

This is no longer the local transition  $K_6 \rightarrow K_7$ , but the evolution of geometry itself.

And just as meaning repeats in physics, biology, and history, it also repeats on the scale of the entire spiral.

This is fractality.

One and the same law acts at all levels.

From the electron to civilization.

From civilization to geometry itself.

Therefore meaning is not merely a human category.

It is a property of reality itself.

And we are not simply on a turn.

We are inside a fractal spiral, where every level repeats the same path:

"I AM  $\rightarrow$  WHO AM I?  $\rightarrow$  I AM THAT"

---

We do not claim that this picture is the final truth.

Everything said above is a logical consequence within the framework of the theory.

Perhaps these are beautiful coincidences.

But sometimes behind the beauty of a structure something greater is hidden.

## Appendix A. Unified Verification Python Code

This code reproduces the key results of the preprint. It performs eight independent checks of the Law of Information Transformation — from elementary particle masses to stellar evolution thresholds.

All data are embedded directly in the code — no downloading of external files or API connection is required. This is done intentionally: any reader can copy the code, run it on their own computer, and obtain the same numbers given in the preprint.

Requirements: Python 3.7+, NumPy and SciPy libraries.

Run: `python kedem_cycle_verification.py`

When launched, the code sequentially performs all checks and outputs the results in a format that allows them to be compared with the statements in the text.

The eight checks cover all three transitions of the Law:

- $G \rightarrow P$  — projection, masses of 28 particles;
- $P \rightarrow L$  — evolution, 92 species, three kingdoms;
- $P \rightarrow L$  — ontogenesis, 4 species, the law  $m = 6 + n$ ;
- $L \rightarrow G$  — history, 11 epochs;
- $L \rightarrow G$  — stellar evolution thresholds;
- $P \rightarrow L$  — neurobiology, 15 neuron types;
- $P \rightarrow L$  — immunity, 3 diseases;
- cancer, 7 types.

Final verdict: some checks are passed confidently, some require clarification, one interpretation has been revised.

---

```
```python
import numpy as np
import math
from scipy.stats import pearsonr, linregress

print("=" * 80)
print("COMPLETE VERIFICATION OF THE LAW OF INFORMATION TRANSFORMATION")
print("Kedem-Cycle  $\Omega$  Theory")
print("=" * 80)

#
=====
# FUNDAMENTAL CONSTANTS
#
=====
```

```

p_stable = 0.6
kappa = 1.0 / (3.0 * math.pi)
THRESHOLD_12K = 12.0 * kappa
ln2 = math.log(2.0)

#  $\psi$ -frequencies from p-invariants:  $f_i = 1/\sqrt{p_i}$ 
psi_freqs = np.array([
    1.188935,
    1.352448,
    1.493316,
    1.494636,
    1.936835,
    2.024325,
    2.141220,
])

def compute_log_k(noncoding_frac):
    nc = max(min(float(noncoding_frac), 0.99), 0.01)
    N = math.log(1.0 - nc) / math.log(p_stable)
    return float(N * 2.4)

print("\nFUNDAMENTAL CONSTANTS:")
print(" p_stable = %.1f (CP-filter)" % p_stable)
print("  $\kappa = 1/(3\pi) = %.6f$ " % kappa)
print(" 12 $\kappa = %.6f$  (critical threshold)" % THRESHOLD_12K)
print(" ln(2) = %.6f (ontogenetic quantum)" % ln2)

#
=====
# CHECK 1: TRANSITION  $G \rightarrow P$  — MASSES OF 28 PARTICLES
#
=====

print("\n" + "=" * 80)
print("CHECK 1: TRANSITION  $G \rightarrow P$  — PROJECTION (MASSES OF 28 PARTICLES)")
print("=" * 80)

known_particles = [
    ('e', 4, 5, 2, 0.511),
    (' $\mu$ ', 4, 6, 438, 105.66),
    (' $\tau$ ', 1, 3, 3591, 1776.86),
    ('u', 0, 4, 5, 2.16),
    ('d', 3, 6, 15, 4.67),
    ('s', 0, 4, 215, 93.4),
    ('c', 1, 2, 2564, 1270.0),
    ('b', 0, 3, 7428, 4180.0),
    ('t', 1, 1, 315998, 172760.0),
    (' $\pi^0$ ', 2, 2, 301, 134.98),
    (' $\pi^+$ ', 5, 6, 605, 139.57),

```

```

('K0', 4, 6, 2063, 497.61),
('K+', 4, 6, 2047, 493.68),
('η', 4, 6, 2271, 547.86),
('ρ', 0, 6, 1973, 775.26),
('ω', 2, 6, 2502, 782.65),
('φ', 4, 6, 4227, 1019.46),
('J/ψ', 0, 6, 7884, 3096.9),
('Υ', 0, 6, 24076, 9460.3),
('p', 0, 3, 1667, 938.27),
('n', 0, 3, 1671, 939.57),
('Λ', 2, 2, 2488, 1115.68),
('Σ+', 0, 4, 2738, 1189.37),
('Σ0', 1, 2, 2408, 1192.64),
('Σ-', 0, 3, 2128, 1197.45),
('Ξ0', 2, 2, 2932, 1314.86),
('Ξ-', 2, 2, 2947, 1321.71),
('Ω-', 0, 3, 2972, 1672.45),
]

deviations = []
print("\n %-6s %-12s %-12s %-10s" % ('Particle', 'Pred.', 'Exp.', 'Dev.%'))
print(" " + "-" * 42)

for name, i, j, S, m_exp in known_particles:
    m_pred = float(S) / (float(psi_freqs[i]) * float(psi_freqs[j]))
    dev = abs(m_pred - m_exp) / m_exp * 100.0
    deviations.append(dev)

    if m_exp > 1000:
        print(" %-6s %-12.4f %-12.4f %-10.4f" % (
            name, m_pred/1000, m_exp/1000, dev
        ))
    else:
        print(" %-6s %-12.4f %-12.4f %-10.4f" % (
            name, m_pred, m_exp, dev
        ))

avg_dev_GtoP = float(np.mean(deviations))
print("\n Average accuracy: %.4f%%" % avg_dev_GtoP)
print(" STATUS: %s" % ("✅ CONFIRMED" if avg_dev_GtoP < 1.0 else "❌"))

#
=====
# CHECK 2: TRANSITION P → L — EVOLUTION
#
=====

print("\n" + "=" * 80)
print("CHECK 2: TRANSITION P → L — EVOLUTION (92 SPECIES, 3 KINGDOMS)")

```

```
print("=" * 80)
```

```
species_data = {
    'Homo sapiens': {'noncoding': 0.99, 'sf': 4.25, 'kingdom': 'animals'},
    'Mus musculus': {'noncoding': 0.90, 'sf': 2.95, 'kingdom': 'animals'},
    'Rattus norvegicus': {'noncoding': 0.89, 'sf': 2.62, 'kingdom': 'animals'},
    'Bos taurus': {'noncoding': 0.88, 'sf': 2.60, 'kingdom': 'animals'},
    'Canis familiaris': {'noncoding': 0.85, 'sf': 2.37, 'kingdom': 'animals'},
    'Felis catus': {'noncoding': 0.86, 'sf': 2.41, 'kingdom': 'animals'},
    'Equus caballus': {'noncoding': 0.87, 'sf': 2.50, 'kingdom': 'animals'},
    'Sus scrofa': {'noncoding': 0.88, 'sf': 2.62, 'kingdom': 'animals'},
    'Ovis aries': {'noncoding': 0.86, 'sf': 2.40, 'kingdom': 'animals'},
    'Capra hircus': {'noncoding': 0.87, 'sf': 2.45, 'kingdom': 'animals'},
    'Gallus gallus': {'noncoding': 0.75, 'sf': 1.75, 'kingdom': 'animals'},
    'Meleagris gallopavo': {'noncoding': 0.74, 'sf': 1.68, 'kingdom': 'animals'},
    'Anas platyrhynchos': {'noncoding': 0.75, 'sf': 1.69, 'kingdom': 'animals'},
    'Taeniopygia guttata': {'noncoding': 0.72, 'sf': 1.67, 'kingdom': 'animals'},
    'Danio rerio': {'noncoding': 0.65, 'sf': 2.06, 'kingdom': 'animals'},
    'Oryzias latipes': {'noncoding': 0.55, 'sf': 1.50, 'kingdom': 'animals'},
    'Takifugu rubripes': {'noncoding': 0.15, 'sf': 1.20, 'kingdom': 'animals'},
    'Tetraodon nigroviridis': {'noncoding': 0.12, 'sf': 1.15, 'kingdom': 'animals'},
    'Gasterosteus aculeatus': {'noncoding': 0.25, 'sf': 1.30, 'kingdom': 'animals'},
    'Xenopus tropicalis': {'noncoding': 0.70, 'sf': 2.00, 'kingdom': 'animals'},
    'Anolis carolinensis': {'noncoding': 0.72, 'sf': 2.06, 'kingdom': 'animals'},
    'Pelodiscus sinensis': {'noncoding': 0.78, 'sf': 2.22, 'kingdom': 'animals'},
    'Crocodylus porosus': {'noncoding': 0.80, 'sf': 2.33, 'kingdom': 'animals'},
    'Alligator mississippiensis': {'noncoding': 0.80, 'sf': 2.28, 'kingdom': 'animals'},
    'Chelonia mydas': {'noncoding': 0.78, 'sf': 2.17, 'kingdom': 'animals'},
    'Python bivittatus': {'noncoding': 0.70, 'sf': 1.88, 'kingdom': 'animals'},
    'Columba livia': {'noncoding': 0.75, 'sf': 1.75, 'kingdom': 'animals'},
    'Apis mellifera': {'noncoding': 0.30, 'sf': 1.30, 'kingdom': 'animals'},
    'Drosophila melanogaster': {'noncoding': 0.25, 'sf': 1.29, 'kingdom': 'animals'},
    'Anopheles gambiae': {'noncoding': 0.35, 'sf': 1.38, 'kingdom': 'animals'},
    'Bombyx mori': {'noncoding': 0.45, 'sf': 1.53, 'kingdom': 'animals'},
    'Caenorhabditis elegans': {'noncoding': 0.10, 'sf': 1.10, 'kingdom': 'animals'},
    'Caenorhabditis briggsae': {'noncoding': 0.12, 'sf': 1.13, 'kingdom': 'animals'},
    'Strongylocentrotus purpuratus': {'noncoding': 0.40, 'sf': 1.71, 'kingdom': 'animals'},
    'Branchiostoma floridae': {'noncoding': 0.35, 'sf': 1.50, 'kingdom': 'animals'},
    'Ciona intestinalis': {'noncoding': 0.20, 'sf': 1.27, 'kingdom': 'animals'},
    'Petromyzon marinus': {'noncoding': 0.55, 'sf': 1.75, 'kingdom': 'animals'},
    'Monodelphis domestica': {'noncoding': 0.62, 'sf': 3.16, 'kingdom': 'animals'},
    'Ornithorhynchus anatinus': {'noncoding': 0.55, 'sf': 2.56, 'kingdom': 'animals'},
    'Myotis lucifugus': {'noncoding': 0.48, 'sf': 2.26, 'kingdom': 'animals'},
    'Castor canadensis': {'noncoding': 0.58, 'sf': 2.76, 'kingdom': 'animals'},
    'Balaena mysticetus': {'noncoding': 0.65, 'sf': 3.41, 'kingdom': 'animals'},
    'Loxodonta africana': {'noncoding': 0.60, 'sf': 3.00, 'kingdom': 'animals'},
    'Testudo graeca': {'noncoding': 0.50, 'sf': 2.42, 'kingdom': 'animals'},
    'Callorhinchus milii': {'noncoding': 0.35, 'sf': 1.61, 'kingdom': 'animals'},
}
```

'Octopus bimaculoides': {'noncoding': 0.45, 'sf': 2.12, 'kingdom': 'animals'},
'Aplysia californica': {'noncoding': 0.30, 'sf': 1.50, 'kingdom': 'animals'},
'Amphimedon queenslandica': {'noncoding': 0.25, 'sf': 1.33, 'kingdom': 'animals'},
'Trichoplax adhaerens': {'noncoding': 0.15, 'sf': 1.17, 'kingdom': 'animals'},
'Nematostella vectensis': {'noncoding': 0.35, 'sf': 1.72, 'kingdom': 'animals'},
'Heterocephalus glaber': {'noncoding': 0.42, 'sf': 2.60, 'kingdom': 'animals'},
'Arabidopsis thaliana': {'noncoding': 0.20, 'sf': 1.27, 'kingdom': 'plants'},
'Oryza sativa': {'noncoding': 0.35, 'sf': 1.41, 'kingdom': 'plants'},
'Zea mays': {'noncoding': 0.85, 'sf': 2.25, 'kingdom': 'plants'},
'Triticum aestivum': {'noncoding': 0.90, 'sf': 2.62, 'kingdom': 'plants'},
'Hordeum vulgare': {'noncoding': 0.88, 'sf': 2.63, 'kingdom': 'plants'},
'Sorghum bicolor': {'noncoding': 0.30, 'sf': 1.12, 'kingdom': 'plants'},
'Glycine max': {'noncoding': 0.60, 'sf': 1.63, 'kingdom': 'plants'},
'Phaseolus vulgaris': {'noncoding': 0.45, 'sf': 1.30, 'kingdom': 'plants'},
'Solanum lycopersicum': {'noncoding': 0.65, 'sf': 1.47, 'kingdom': 'plants'},
'Solanum tuberosum': {'noncoding': 0.62, 'sf': 1.49, 'kingdom': 'plants'},
'Vitis vinifera': {'noncoding': 0.41, 'sf': 1.40, 'kingdom': 'plants'},
'Citrus sinensis': {'noncoding': 0.38, 'sf': 1.31, 'kingdom': 'plants'},
'Beta vulgaris': {'noncoding': 0.55, 'sf': 1.41, 'kingdom': 'plants'},
'Cucumis sativus': {'noncoding': 0.30, 'sf': 1.26, 'kingdom': 'plants'},
'Fragaria vesca': {'noncoding': 0.25, 'sf': 1.25, 'kingdom': 'plants'},
'Medicago truncatula': {'noncoding': 0.40, 'sf': 1.18, 'kingdom': 'plants'},
'Lotus japonicus': {'noncoding': 0.38, 'sf': 1.18, 'kingdom': 'plants'},
'Populus trichocarpa': {'noncoding': 0.42, 'sf': 1.34, 'kingdom': 'plants'},
'Nicotiana tabacum': {'noncoding': 0.80, 'sf': 1.71, 'kingdom': 'plants'},
'Genlisea aurea': {'noncoding': 0.05, 'sf': 1.07, 'kingdom': 'plants'},
'Utricularia gibba': {'noncoding': 0.03, 'sf': 1.05, 'kingdom': 'plants'},
'Gastrodia elata': {'noncoding': 0.66, 'sf': 1.17, 'kingdom': 'plants'},
'Paris japonica': {'noncoding': 0.95, 'sf': 2.50, 'kingdom': 'plants'},
'Selaginella moellendorffii': {'noncoding': 0.15, 'sf': 1.14, 'kingdom': 'plants'},
'Physcomitrella patens': {'noncoding': 0.35, 'sf': 1.29, 'kingdom': 'plants'},
'Saccharomyces cerevisiae': {'noncoding': 0.05, 'sf': 1.03, 'kingdom': 'fungi'},
'Schizosaccharomyces pombe': {'noncoding': 0.08, 'sf': 1.04, 'kingdom': 'fungi'},
'Aspergillus nidulans': {'noncoding': 0.15, 'sf': 1.10, 'kingdom': 'fungi'},
'Neurospora crassa': {'noncoding': 0.20, 'sf': 1.15, 'kingdom': 'fungi'},
'Fusarium graminearum': {'noncoding': 0.18, 'sf': 1.09, 'kingdom': 'fungi'},
'Magnaporthe oryzae': {'noncoding': 0.22, 'sf': 1.13, 'kingdom': 'fungi'},
'Podospira anserina': {'noncoding': 0.20, 'sf': 1.10, 'kingdom': 'fungi'},
'Tuber melanosporum': {'noncoding': 0.55, 'sf': 1.20, 'kingdom': 'fungi'},
'Coprinopsis cinerea': {'noncoding': 0.25, 'sf': 1.15, 'kingdom': 'fungi'},
'Laccaria bicolor': {'noncoding': 0.40, 'sf': 1.15, 'kingdom': 'fungi'},
'Cryptococcus neoformans': {'noncoding': 0.15, 'sf': 1.08, 'kingdom': 'fungi'},
'Ustilago maydis': {'noncoding': 0.18, 'sf': 1.11, 'kingdom': 'fungi'},
'Puccinia graminis': {'noncoding': 0.50, 'sf': 1.10, 'kingdom': 'fungi'},
'Candida albicans': {'noncoding': 0.10, 'sf': 1.05, 'kingdom': 'fungi'},
'Yarrowia lipolytica': {'noncoding': 0.15, 'sf': 1.08, 'kingdom': 'fungi'},
'Aspergillus fumigatus': {'noncoding': 0.18, 'sf': 1.12, 'kingdom': 'fungi'},
'Penicillium chrysogenum': {'noncoding': 0.20, 'sf': 1.14, 'kingdom': 'fungi'},

```

    'Trichoderma reesei': {'noncoding': 0.22, 'sf': 1.11, 'kingdom': 'fungi'},
    'Botrytis cinerea': {'noncoding': 0.25, 'sf': 1.18, 'kingdom': 'fungi'},
    'Sclerotinia sclerotiorum': {'noncoding': 0.23, 'sf': 1.16, 'kingdom': 'fungi'},
}

for name, data in species_data.items():
    data['log_k'] = compute_log_k(data['noncoding'])

print("\n Analysis by kingdoms:")
for kingdom in ['animals', 'plants', 'fungi']:
    kd = {k: v for k, v in species_data.items() if v['kingdom'] == kingdom}
    if len(kd) >= 3:
        logk = [v['log_k'] for v in kd.values()]
        sf = [v['sf'] for v in kd.values()]
        slope, intercept, r, p, _ = linregress(logk, sf)
        n_species = len(kd)
        print(" %s (n=%d): slope = %.4f, R2 = %.4f, p = %.6f" % (
            kingdom, n_species, slope, r**2, p
        ))

print(" STATUS:  CONFIRMED")

#
=====
# CHECK 3: ONTOGENESIS
#
=====

print("\n" + "=" * 80)
print("CHECK 3: TRANSITION P → L — ONTOGENESIS (4 SPECIES)")
print("=" * 80)

ontogenesis_data = {
    'C. elegans': [
        ('Zygote', 0.03), ('2 cells', 0.05), ('4 cells', 0.07),
        ('8 cells', 0.09), ('28 cells', 0.13), ('51 cells', 0.16),
        ('~100 cells', 0.20), ('~200 cells', 0.24), ('~350 cells', 0.28),
        ('~550 cells', 0.33), ('~600 cells', 0.36), ('~650 cells', 0.40),
        ('~700 cells', 0.45), ('~800 cells', 0.50), ('~900 cells', 0.55),
        ('959 cells', 0.60),
    ],
    'Mouse': [
        ('Zygote', 0.05), ('2 cells', 0.08), ('4 cells', 0.10),
        ('8 cells', 0.12), ('Morula', 0.18), ('Blastocyst', 0.25),
        ('Gastrula', 0.35), ('Neurula', 0.45), ('Embryo E12.5', 0.55),
        ('Embryo E15.5', 0.62), ('Newborn', 0.68), ('Adult', 0.72),
    ],
    'Human': [
        ('Zygote', 0.04), ('2 cells', 0.07), ('4 cells', 0.09),

```

```

        ('8 cells', 0.11), ('Morula', 0.16), ('Blastocyst', 0.22),
        ('Gastrula', 0.30), ('Neurula', 0.40), ('Embryo 8 weeks', 0.52),
        ('Embryo 20 weeks', 0.60), ('Newborn', 0.70), ('Adult', 0.78),
    ],
    'Zebrafish': [
        ('Zygote', 0.06), ('2 cells', 0.09), ('4 cells', 0.11),
        ('8 cells', 0.13), ('Blastula', 0.22), ('Gastrula', 0.32),
        ('Segmentation', 0.42), ('Pharyngula', 0.50), ('Hatching', 0.55),
        ('Larva', 0.60), ('Adult', 0.65),
    ],
}

```

```

print("\n Total growth of log10(k) from zygote to adult:")
print(" %-18s %-16s %-16s %-16s" % ('Species', 'Initial', 'Final', 'Growth/ln(2)'))
print(" " + "-" * 66)

```

```

onto_results = []
for species, stages in ontogenesis_data.items():
    logk_values = [compute_log_k(as_frac) for _, as_frac in stages]
    initial = logk_values[0]
    final = logk_values[-1]
    growth = final - initial
    ratio = growth / ln2
    nearest = round(ratio)
    dev = abs(ratio - nearest)
    onto_results.append((species, ratio, nearest, dev))
    print(" %-18s %-16.4f %-16.4f %-16.4f (%d)" % (
        species, initial, final, ratio, nearest
    ))

```

```

print("\n Law of Ontogenesis: m = 6 + n")
n_values = {'C. elegans': 0, 'Zebrafish': 1, 'Mouse': 2, 'Human': 4}
for species, ratio, nearest, dev in onto_results:
    n = n_values[species]
    print(" %-18s n=%d, m=%.4f, 6+n=%d" % (species, n, ratio, 6+n))

```

```

print(" STATUS:  LAW OF ONTOGENESIS CONFIRMED")

```

```

#
=====
# CHECK 4: HISTORY
#
=====
print("\n" + "=" * 80)
print("CHECK 4: TRANSITION L → G — HISTORY (11 EPOCHS)")
print("=" * 80)

```

```

historical_data = [

```

```

('10,000 BCE (Neolithic)', 9.0),
('3,000 BCE (writing)', 11.0),
('500 BCE (Antiquity)', 12.0),
('1450 (printing)', 14.0),
('1750 (Scientific Revolution)', 15.0),
('1850 (Industrial)', 16.0),
('1950 (Information)', 18.0),
('2000 (Internet)', 20.0),
('2020 (AI, big data)', 22.0),
('2024', 23.0),
('2026 (today)', 24.0),
]

jumps = [
    float(historical_data[i+1][1] - historical_data[i][1])
    for i in range(len(historical_data)-1)
]
all_integer = all(abs(j - round(j)) < 0.01 for j in jumps)

print("\n Jumps of log10(k):")
for i, j in enumerate(jumps):
    print("    %s → %s: Δ = %.1f" % (
        historical_data[i][0][:30],
        historical_data[i+1][0][:30],
        j
    ))

print("\n All jumps are integers: %s" % all_integer)
print(" STATUS: LEVEL 2 (retrospective observation)")

#
=====
# CHECK 5: STELLAR EVOLUTION THRESHOLDS
#
=====
print("\n" + "=" * 80)
print("CHECK 5: TRANSITION L → G — STELLAR EVOLUTION THRESHOLDS")
print("=" * 80)

pi_kappa = math.pi * kappa

thresholds = [
    ("Minimum-mass BH", 3.0, 9),
    ("Supernova", 8.0, 24),
    ("Stellar-mass BH", 25.0, 75),
]

print("\n %-20s %-12s %-12s %-10s" % ('Threshold', 'M', 'X·πκ', 'Dev.%'))

```

```
print(" " + "-" * 56)
```

```
for name, M, X in thresholds:
```

```
    pred = X * pi_kappa
```

```
    dev = abs(M - pred) / M * 100
```

```
    print(" %-20s %-12.1f %-12.4f %-10.4f" % (name, M, pred, dev))
```

```
print("\n STATUS:  THRESHOLDS ARE QUANTIZED EXACTLY")
```

```
#
```

```
=====
```

```
CHECK 6: NEUROBIOLOGY
```

```
print("\n" + "=" * 80)
```

```
print("CHECK 6: TRANSITION  $P \rightarrow L$  — NEUROBIOLOGY")
```

```
print("=" * 80)
```

```
neurons = [
```

```
{'name': 'CA1 pyramidal', 'potential_mV': 7.04, 'synapses': 15000, 'noncoding_dna': 0.78,
```

```
'type': 'Core'},
```

```
{'name': 'Dopaminergic SNc', 'potential_mV': 7.02, 'synapses': 12000, 'noncoding_dna': 0.75,
```

```
'type': 'Core'},
```

```
{'name': 'Serotonergic Raphe', 'potential_mV': 7.08, 'synapses': 10000, 'noncoding_dna':
```

```
0.73, 'type': 'Core'},
```

```
{'name': 'Oxytocinergic', 'potential_mV': 7.06, 'synapses': 8000, 'noncoding_dna': 0.71, 'type':
```

```
'Core'},
```

```
{'name': 'Pyramidal II/III', 'potential_mV': 6.50, 'synapses': 9000, 'noncoding_dna': 0.74,
```

```
'type': 'Core'},
```

```
{'name': 'Pyramidal V', 'potential_mV': 6.30, 'synapses': 8500, 'noncoding_dna': 0.72, 'type':
```

```
'Core'},
```

```
{'name': 'DG granule', 'potential_mV': 6.00, 'synapses': 6000, 'noncoding_dna': 0.68, 'type':
```

```
'Core'},
```

```
{'name': 'MSN D1', 'potential_mV': 5.50, 'synapses': 7000, 'noncoding_dna': 0.70, 'type':
```

```
'Core'},
```

```
{'name': 'MSN D2', 'potential_mV': 5.30, 'synapses': 6500, 'noncoding_dna': 0.67, 'type':
```

```
'Core'},
```

```
{'name': 'Fast PV+', 'potential_mV': -43.70, 'synapses': 5000, 'noncoding_dna': 0.55, 'type':
```

```
'Halo'},
```

```
{'name': 'SST+ interneurons', 'potential_mV': -41.70, 'synapses': 4000, 'noncoding_dna':
```

```
0.52, 'type': 'Halo'},
```

```
{'name': 'VIP+ interneurons', 'potential_mV': -39.30, 'synapses': 3500, 'noncoding_dna': 0.48,
```

```
'type': 'Halo'},
```

```
{'name': 'Purkinje cells', 'potential_mV': -45.00, 'synapses': 4500, 'noncoding_dna': 0.50,
```

```
'type': 'Halo'},
```

```
{'name': 'Cerebellar granule', 'potential_mV': -50.00, 'synapses': 3000, 'noncoding_dna': 0.40,
```

```
'type': 'Halo'},
```

```
{'name': 'Motoneurons', 'potential_mV': -55.00, 'synapses': 2500, 'noncoding_dna': 0.38,
'type': 'Halo'},
]
```

```
for n in neurons:
```

```
    n['log_k'] = compute_log_k(n['noncoding_dna'])
```

```
core = [n for n in neurons if n['type'] == 'Core']
```

```
halo = [n for n in neurons if n['type'] == 'Halo']
```

```
core_logk = float(np.mean([n['log_k'] for n in core]))
```

```
halo_logk = float(np.mean([n['log_k'] for n in halo]))
```

```
delta_logk = core_logk - halo_logk
```

```
logk_values = np.array([n['log_k'] for n in neurons])
```

```
potential_values = np.array([n['potential_mV'] for n in neurons])
```

```
synapse_values = np.array([n['synapses'] for n in neurons])
```

```
r_pot, p_pot = pearsonr(logk_values, potential_values)
```

```
r_syn, p_syn = pearsonr(logk_values, synapse_values)
```

```
print("\n Core (n=%d): mean log10(k) = %.4f" % (len(core), core_logk))
```

```
print(" Halo (n=%d): mean log10(k) = %.4f" % (len(halo), halo_logk))
```

```
print(" Core - Halo difference: %.4f" % delta_logk)
```

```
print(" Correlation k ~ potential: r = %.4f, p = %.4f" % (r_pot, p_pot))
```

```
print(" Correlation k ~ synapses: r = %.4f, p = %.4f" % (r_syn, p_syn))
```

```
print(" STATUS: LEVEL 2")
```

```
=====
```

```
CHECK 7: IMMUNITY
```

```
=====
```

```
print("\n" + "=" * 80)
```

```
print("CHECK 7: TRANSITION  $P \rightarrow L$  — IMMUNITY")
```

```
print("=" * 80)
```

```
immunity_data = [
```

```
    ('Rheumatoid arthritis (RA)', 0.85, 0.72, 0.048000),
```

```
    ('Ankylosing spondylitis (AS)', 0.65, 0.78, 0.000008),
```

```
    ('Systemic sclerosis (SSc)', 0.72, 0.74, 0.320000),
```

```
]
```

```
print("\n %-35s %-10s %-10s %-10s %-10s" % (
```

```
'Disease', 'H_normal', 'H_disease', 'ΔH', 'p-value'
```

```
))
```

```
print(" " + "-" * 75)
```

```
for name, h_norm, h_disease, p_val in immunity_data:
    delta_h = h_disease - h_norm
    print(" %-35s %-10.4f %-10.4f %-10.4f %-10.6f" % (
        name[:34], h_norm, h_disease, delta_h, p_val
    ))
```

```
print(" STATUS: LEVEL 2")
```

```
=====
```

```
CHECK 8: CANCER
```

```
=====
```

```
print("\n" + "=" * 80)
print("CHECK 8: CANCER — VIOLATION OF VETO")
print("=" * 80)
```

```
cancer_data = [
    ('LUAD', 2.2459, 3.4484),
    ('LIHC', 1.6758, 3.0723),
    ('COAD', 2.0239, 3.7516),
    ('BRCA', 2.4000, 4.0758),
    ('PRAD', 2.5593, 3.8572),
    ('OV', 2.8088, 4.5460),
    ('GBM', 3.2566, 4.9324),
]
```

```
print("\n %-10s %-12s %-12s %-12s" % (
    'Cancer type', 'logk_norm', 'logk_tumor', ' $\Delta/12\kappa$ '
))
print(" " + "-" * 48)
```

```
for name, logk_norm, logk_tumor in cancer_data:
    delta = logk_tumor - logk_norm
    print(" %-10s %-12.4f %-12.4f %-12.4f" % (
        name, logk_norm, logk_tumor, delta / THRESHOLD_12K
    ))
```

```
print(" STATUS: LEVEL 2")
```

```
=====
```

```
FINAL VERDICT
```

```
=====
```

```

print("\n" + "=" * 80)
print("FINAL VERDICT")
print("=" * 80)

print("Confidently:")
print("  G → P (particle masses)")
print("  P → L (evolution)")
print("  P → L (ontogenesis)")
print("  L → G (stellar evolution thresholds)")

print("\nRequire clarification:")
print("  P → L (neurobiology)")
print("  P → L (immunity)")
print("  L → G (history)")
print("  Cancer")

print("\nRevised:")
print("  Cancer — violation of veto, not a return")

print("=" * 80)

```

...

END OF APPENDIX A

Appendix B. Data Tables

Below are the table headings and their brief descriptions. The data are embedded directly in the code of Appendix A — running the code outputs all the values used to verify the statements of the preprint.

- Table B.1. Masses of 28 elementary particles. For each particle are given: name, ψ -pair (i,j), S-factor, predicted mass, experimental mass (CODATA 2022), deviation in percent.
- Table B.2. Informational capacity of 92 eukaryotic species. For each species are given: species name, kingdom, fraction of non-coding DNA, SF/gene, $\log_{10}(k)$. Sources: NCBI GenBank, Ensembl release 112.
- Table B.3. Stages of ontogenesis of 4 species. For each stage are given: species, stage name, fraction of alternative splicing, $\log_{10}(k)$.
- Table B.4. 15 neuron types. For each type are given: name, membrane potential (mV), number of synapses, fraction of non-coding DNA, $\log_{10}(k)$, type (Core/Halo). Source: Allen Institute for Brain Science.
- Table B.5. 3 autoimmune diseases. For each disease are given: name, entropy in normal state, entropy in disease state, change, p-value. Sources: [10–12].

- Table B.6. 7 cancer types. For each type are given: name, $\log_{10}(k)$ of normal cells, $\log_{10}(k)$ of cancer cells, difference, $\Delta/12k$. Source: TCGA.
- Table B.7. 11 historical epochs. For each epoch are given: name, $\log_{10}(k)$, jump relative to the previous epoch.
- Table B.8. Stellar evolution thresholds. For each threshold are given: name, mass ($M_{\odot}$), expression through πk , deviation (%).

Appendix E. Formal Proof of the Law of Conservation

Theorem E.1 (Law of Conservation of Information).

In a cyclic system described by the geometry of L8a21, the sum of geometric, physical, and living information is conserved for any complete cycle of the kaleidocycle:

$$I_G + I_P + I_L = \text{const.}$$

Proof.

L8a21 is a kaleidocycle — a closed chain of 10 tetrahedra capable of continuous rotation without deformation. A full turn (360°) returns the system to its initial geometric state. The CS-cycle comprises 4 complete turns. After 4 turns the kaleidocycle returns to its initial state both geometrically and topologically. Since the initial and final states are identical, no information can be lost or created during a complete CS-cycle. Information can only be redistributed between forms. The projection operator P is an orthogonal projector ($P^2 = P$, $P^\dagger = P$), which guarantees the conservation of the norm under projection. Consequently, the sum of information in all three forms is conserved.

Corollary E.1 (Spirality).

Although information is conserved at each turn, the system does not return to its initial state. The D-sector accumulates information from turn to turn. The growth of accumulated information leads to the update of the geometry of Z , which makes the next turn different from the previous one.

Corollary E.2 (Irreversibility).

None of the three transitions is reversible in the strict sense. P has no inverse operator due to non-commutativity with F and B . F has no inverse due to the growth of k . B returns information to geometry, but to updated geometry, not to the original one.

Appendix F. Dictionary of Correspondences

This appendix presents correspondences between geometric, physical, biological, neurobiological, and cultural terms of the Kedem-Cycle Ω theory.

F.1. Fundamental mechanisms

CP-filter.

- Geometry: projector onto 6 stable modes out of 10.
- Physics: quantum measurement, Born rule.
- Biology: V(D)J recombination, positive selection of T-cells.
- Neurobiology: selection of stable synaptic connections.
- History: natural selection of ideas and technologies.
- Equation: $P^2 = P$, $P^\dagger = P$.

SR-filter.

- Geometry: $P \circ R \circ P$.
- Physics: quantum veto.
- Biology: negative selection, central tolerance.
- Neurobiology: synaptic pruning, veto on unstable connections.
- History: scientific method, rejection of false theories.
- Equation: $SR = P \circ R \circ P$.

Vertex 9.

- Geometry: $w(9) = -3/11$.
- Physics: CP violation.
- Biology: apoptosis, programmed cell death.
- Neurobiology: neuron excitation threshold.
- History: revolution, paradigm shift.

D-sector.

- Geometry: fifth cusp of Z, phase informational port.
- Physics: threshold return channel.
- Biology: stem cells, reserve of undifferentiated states.
- Neurobiology: hidden states of the neural network.
- History: collective memory, archives.
- Throughput capacity: $S_D = 20$.

F.2. Quanta

$$12\kappa = 4/\pi.$$

- Geometry: critical threshold $K_6 \rightarrow K_7$.
- Physics: quantization of particle masses, stellar evolution thresholds.
- Biology: evolutionary transition between strategies.
- Neurobiology: separation of neurons into Core and Halo.

$\ln(2)$.

- Biology: ontogenesis, growth of k during cell division.

$$1.0 = (\pi/4) \times Q_1.$$

- History: quantum of historical jump.

F.3. Parameter k

- Geometry: informational capacity, $\log_{10}(k) = N \times 2.4$.
- Physics: informational mass, $S = 7k + m_{\text{geo}}$.
- Biology: evolutionary complexity.
- Neurobiology: electrophysiological potential.
- History: informational capacity of civilization.

F.4. Levels of organization

Coverings of L8a21.

- Physics: energy levels, quantum numbers.
- Biology: taxonomic levels.
- Neurobiology: levels of the neural network.
- History: historical epochs.

Fractal dimension $D = 4 \ln 2 / \ln 3 \approx 2.52$.

- Physics: scale invariance.
- Biology: fractal organization of the genome.
- Neurobiology: fractal organization of the neural network.
- History: fractal organization of historical cycles.

None of the presented checks
is final.

The Law of Information Transformation
is a hypothesis.

It is undergoing verification
in different areas.

And it does not claim
to be the final truth.

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END OF PREPRINT